

Section A

Database : Organised collection of data.

× Banks (FRIB, ABSP and etc)

* Online Retail (Takealot, amazon, Temu)

2. Data marts : A subset of a data warehouse, focused on a specific functional area (e.g., sales, finance).

= A company would use it because it has, faster access, faster response, lower cost, more focused.

3. Data lake : gathers and stores data in its original format in a central location.

Difference : Warehouse stores structured, cleaned data. Lake stores raw, all types.

Business prefers a lake when it has big data, unstructured data, or when data scientists want to do machine learning and don't know yet what data they need.

4. Data warehouse : A central repository that collects data from multiple sources to support business intelligence and decision-making.

Regular data	Data warehouse
fast, small, current data	slow, large, historical data
Data is always changing	Data is stable, cleaned.
used by cashiers, customers	used by managers, analysts

5 Structured data : data does not fit in tables but has some tags or markers that organize it. it has no fixed schema but is flexible.

Example : An employees table in MySQL or Excel sheet with Columns: ID, Name, Salary, Department, e.g. 1, John, 60000, IT.

Semi-Structured data : This data does not fit in

5. Structured data : This is data that is highly organized in a fixed format, like tables with rows and columns. It is easy to search using SQL. It has a clear schema.

Example : An employees table in MySQL : ID, Name, Salary, Department e.g.,
John, 50000, IT.

2. Semi-Structured Data : This data does not fit in tables but has some tags or markers that organize it. It has no fixed schema but is flexible.

Example : A JSON file or XML file. e.g.
{ "name" : "John", "city" : "Johannesburg" } or an Email with Subject and body.

3. Unstructured Data : This is data with no Model at all, messy and hard to search. It is the biggest part of data in the world.

Example : Whatsapp voice note, Instagram video, CCTV footage or a photo.

6. Data Modeling : is making a blueprint of how data will be stored and linked before building the database.

Important

1. Saves time and money
2. Prevents duplication and errors

Two types

1. Conceptual Model - High-level, simple view for business, shows entities only e.g. Customer places order.
2. Logical Model - Detailed view for developers - shows tables, columns, primary key, foreign key.

Q7 Data Mining is finding hidden patterns from large data.
used in business ° To understand customers and predict sales.

Examples °

1. Takealot - recommends products based on what you bought before.
2. Banks - detects fraud when unusual spending happens.

Q8 DBMS is software that helps you create, manage and control a database.

1. MySQL - Free, fast, relational. Use ° Small websites, e.g. Takealot stores.
2. Oracle - Powerful, Secure, expensive. Use ° Big banks like FNB for large transactions.
3. PostgreSQL - Free, advanced, open-source. Use ° Data analytics and apps that need complex data.
- c. MongoDB - NoSQL, stores unstructured data like JSON. Use ° Instagram, TikTok for storing videos/comments.

Q9 ETL = Extract, Transform, Load - process of moving data from different sources into one data warehouse for analysis.

- ° Extract ° Gather data in suitable format
- ° ~~Extract~~ ° Transform ° Select needed data and put in correct format.
- ° Load ° Store in the warehouse (clean data).

Important

without ETL, data is dirty and scattered. ETL gives one clean, trusted place to do reports and make business decisions.

Q10

1. CSV: Single table with rows and columns.

2. JSON: Stores data as key-value pairs.

flexible. Use: APIs, web apps

unstructured data like Social Media

Comments.

3. Parquet - Column-based highly compressed, very fast. Use: big data analytics, data warehouses, Best for analytics.

4. Avro - Row-based, includes schema, good for streaming. Use: Kafka, real-time data streaming.

When to use which?

• Small report = CSV

• API/web = JSON

• Large analytics = Parquet

• Streaming = Avro

Section 13

Q11 Artificial Intelligence (AI)

AI is making computers think and learn like humans to do tasks.

Narrow AI (Weak AI) : AI made for one specific task. It exists now.

- Example : Siri, Netflix, Music recommendations.

General AI (Strong AI) : AI that can think and do any human task like a human. It doesn't fully exist yet.

- Example : A human-like robot that can cook, drive, and do your job all at once.

Q12 Machine Learning (ML)

ML is when computers learn from data instead of being given rules.

- Traditional : You give rules + data = answer
- ML : You give data + answer = Computer learns rules itself.

3 Types :

1. Supervised : Learns from labelled data. Ex : Emails spam filter (you show it what's spam/not spam).
2. Unsupervised : Finds patterns in unlabeled data. Ex : Shoprite grouping customers by buying habits.
3. Unsupervised : Finds patterns in unlabeled data. Ex : Shoprite grouping customers by buying habits.

Q13. AI > ML > Deep Learning.

AI is the big idea. ML is a type of AI. Deep Learning is a type of ML.

Deep Learning - Uses big Neural Networks to Learn from huge data.

Neural Network: works like a human brain. has layers of connected nodes (neurons). Data goes in layers find patterns to give Output.

Ex: Face ID on your phone.

Q14 Natural Language Processing (NLP)?

NLP is how computers understand, read and speak human language.

Examples:

1. Auto-correct / Grammarly - fixes your typing.
2. Google Translate - translates languages.
3. Siri / Whatsapp voice-to-text - understands your voice.

Q15 Large Language Model

LLM = is a huge AI Model trained on massive text to understand and generate human language.

Difference

- Standard ML: Small, does 1 task (e.g., predict sales).
- LLM: Huge, trained on billions of words, can do many language tasks.

2. Examples:

1. ChatGPT (Open AI) - for chat, writing, coding.
2. Gemini (Google) - for search, writing, answering questions.

Q16 Generative A-I is AI that CREATES new Content (Text, images, music) instead of just analyzing it.

How it works: Learns patterns from huge data, then predicts what comes next to generate something new.

3 Tools.

1. ChatGPT - Generates text / essays / code.
2. Midjourney / Dall-E - Generates images from text.
3. Suno - Generates music from text.

Q17 Training Dataset: Data used to teach the AI Model.

Why quality is important: Bad data = Bad AI. Model learns whatever you show it.

If data is biased / incorrect:

1. Biased results - e.g., Only shows male doctors if trained only on male photos.
2. Wrong predictions - AI will make mistakes.
3. Unfair / Harmful - Can discriminate against people.

Q18 Computer Vision: is teaching Computers to see and understand images/videos like humans

How they see: Camera captures images as pixels then AI finds patterns.

2 Cases:

1. Face ID on phones - unlocks your phone by seeing your face
2. Self-driving cars - sees road, cars, people to drive safely.

Section C: API, Tools & Developer Concepts.

Q19 API (Application programming Interface)

It's a way for 2 apps to talk to each other.

analogy: like a waiter, you (APP1) order

waiter (API) kitchen (APP2) bring food back.

Why important:

1. Apps can share data easily.
2. Builds bigger systems fast without rebuilding everything.

Q20

API key: A Secret password/code to use an API.

purpose: To identify who you are and track usage/billing.

How to protect: store in environment file (.env), NOT in code / GitHub.

if exposed: Hackers can steal it, use your account, run up big bills.

Q21

API REST = Representation State Transfer. It's a standard way to build APIs using HTTP.

4. Main Methods.

1 Get - Get / read data

2 Post - Create new data

3 Put - update existing data.

4 Delete - Remove data.

Q22 Json (JavaScript Object Notation)?

= A simple text format to send data.

Light-weight and easy to read.

Why used? Both humans and machines can read it. works with all languages.

Ex^e Json - code

```
{ "name": "John",  
  "age": 24,  
  "is Student": true  
}
```

Q23. Prompt Engineering - Writing clear instructions to get best results from AI.

Why Important? Better prompt = Better answer. Bad prompt = useless answer.

Examples:

1. Poor: Write about AI

Better: Write a 100-word explanation of AI for a 10-year-old.

2. Poor: fix my code

Better: My Python code gives error X. Here is code: [Code]. Explain and fix it.

Q24 MAP = Takes a list applies a function to every item, returns a new list.

Python code

```
nums = [1, 2, 3]
```

```
new_nums = list(map(lambda x: x * 2, nums))
```

Results: [2, 4, 6]

Q25 Cloud Computing = using internet to rent computing instead of buying servers.

1. IaaS - Rent Infrastructure. Ex: AWS EC2
2. PaaS - Rent platform to build apps. Ex: Heroku
3. SaaS - Ready-to-use Software. Ex: Gmail.

Q26 Version Control - System to track changes in files.

Sit: Most popular VCS. Lets teams work together, go back to old versions.

Commit: Save point of your changes.

Branch: Separate copy to try new work without breaking main code.

Q27 Jupyter Notebook

Interactive document to write code + text + charts together.

Language: Python

Why popular: Run code in chunks, see results instantly, easy to share analysis.

Q28 Python - Easy, readable programming language.

Why popular for AI/Data: Simple syntax, huge libraries, big community.

4 Libraries

1. Pandas - data tables & cleaning
2. NumPy - math on arrays
3. Matplotlib - charts/graphs
4. Scikit-Learn - machine learning.

Section D: Data Ethics, Governance & Careers.

Q29 Data privacy: Protecting personal data from misuse.

Important: Prevents identity theft, harm.

GDPR: EU Law since 2018 giving users control over data.
Applies to all EU/EEA countries + anyone handling EU citizens' data.

Q30 Data Ethics - Moral rules for using data.

1. Facial recognition without consent - violates privacy.
2. Selling health data - exploitation
3. AI hiring that filters women - discrimination.

Q3.1 Data Governance - Managing data quality, security, roles/ownership, policies, compliance.

Q.3.2 Bias in AI

Model makes unfair decision because training data was unbalanced.

Ex: Amazon hiring AI biased against women because trained on male resumes.

Consequence: Discrimination, legal issues.

Q.3.3	Job	Key skill	Tools
Data Analyst	Explain what happened	SQL, Excel Dashboard	Power BI, Tableau
Data Scientist	Predict what will happen	Python, Stats ML	Python, Scikit-Learn
Data Engineer	Build pipeline for data.	ETL, SQL Cloud	AWS, Spark, Airflow
AI Engineer	Build & deploy AI models	Deep Learning MLOps	TensorFlow, PyTorch

Q.3.4 Business Intelligence (BI)

Tools that turn raw data into dashboards for decision.

2. Tools

1. Power BI - cheap, Microsoft, easy reports.

2. Tableau - Strong visuals, drag-drop.

Help by showing KPIs, trends in real time.

Section E

Q35 Most Interesting?

Bias in AI - it's surprising how unfair data creates unfair AI.

Q3.6 Which most difficult?

Data Governance - lots of abstract policies. Uses youtube + IBM articles to understand.

Q.3.7 Data / AI

x Data will make our work meaningful and duplication
x AI will help make work easier and easy to access

Examples of DATA

helps me gather information safely and organised and avoiding making many same work.

AI

* IT helps us study and make examples, instead of reading the whole book or 2 pages it refers you to what you are looking for. (Save time)

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